

MSST Transformer: A Novel Multimodal Spatial-Spectral-Temporal Transformer for Time-series Hyperspectral Imaging in Plant Growth Modeling

Michael Hua

Cranbrook Kingswood School
Bloomfield Hills, Michigan, United States

Contents

1	Introduction	3
2	Related Work	5
2.1	Computer Vision-based Sensing of Plant's Status	5
2.1.1	RGB Imaging-based Sensing	5
2.1.2	Hyperspectral Imaging-based Sensing	5
2.2	AI Techniques for Analyzing Plant Growth under Environmental Conditions	6
2.2.1	Convolutional Neural Network	6
2.2.2	Transformer	7
3	Novel Spatial-Spectral-Temporal Transformer for Multimodal Modeling of Time-series Hyperspectral Images	8
3.1	Hyperspectral Imaging and Reflectance Computation	8
3.2	Spatial-Spectral Vision Transformer for Hyperspectral Image Embedding	9
3.3	Novel Spatial-Spectral-Temporal Transformer for Multimodal Time-Series Modeling of Plant Growth	12
3.3.1	Spatial-Spectral-Temporal Transformer	12
3.3.2	Cross-Attention Exploration of Multimodal Relationship	13
3.3.3	Loss Function	13
4	System Development and Experiments	14
4.1	Experiment Condition and Setup	14
4.2	Hyperspectral Image and Growth Data Collection	15
4.3	Algorithm Setup, Implementation and Training Details	16
4.4	Quantitative Metrics for Performance Evaluation	17
4.5	Evaluation and Comparison of the Multimodal Spatial-Spectral-Temporal Transformer	18
4.6	Model Analysis and Ablation Study	18
4.7	Improved Light Treatment Strategy	18
5	Conclusion	19
6	References	19

Abstract

Controlled environmental agriculture provides innovative methods for cultivating plants to address challenges like food security, environmental sustainability, and urban agriculture. Precision control of environmental conditions has the potential to significantly improve plant growth. Accurate modeling of the interaction between the plant and its environment is essential for implementing precision control. Hyperspectral imaging has high potential for this purpose because it can capture detailed spectral information across a wide range of wavelengths from plants. However, the higher dimensionality, larger sizes, and complex nature of hyperspectral images pose a significant challenge if used for temporal studies of plant growth. Subtle variations in plant properties and physiological changes need to be identified and correlated to provide accurate plant growth assessment. This paper introduces a novel multimodal spatial-spectral-temporal transformer designed to analyze the space, spectrum, and time domains of the hyperspectral images. The transformer is trained on hyperspectral images of growing plants, corresponding light treatments, and final biomass outcomes to learn a statistical model which captures the intrinsic relationship between plant growth patterns and environmental factors. Specifically, the spatial-spectral transformer divides the input image into 3D hyperspectral patches and utilizes a novel 3D-aware positional encoding and self-attention mechanisms to capture global dependencies among patches. Subsequently, the temporal transformer models time-variant spatial-spectral representations and their long-range relationships. Finally, the multimodal cross-attention mechanism explores the interaction between environmental features and spatial-spectral-temporal representations from hyperspectral plant images, resulting in an accurate plant growth model. Comprehensive experiments demonstrate the effectiveness and superiority of the proposed network.

1 Introduction

Vertical farming systems offer effective solutions for cultivating crops, which have the potential to address critical challenges in food security, environmental sustainability, and urban agriculture [Van Delden et al., 2021, Kaiser et al., 2024]. Plant growth conditions and physiological changes demand precise environmental control. For example, optimizing light treatment, one of the most important factors, can produce improved results for plant growth [Massa et al., 2008, Demotes-Mainard et al., 2016]. Accurate modeling of plant growth in response to these environmental factors is vital for achieving precision agriculture. In current practice, monitoring and modeling is often conducted through biologists’ manual assessments, such as using LiCor LI6800 [Meeker et al., 2021], which is extremely time-consuming. The increasing demand for online plant growth monitoring has driven the development of automated, high-throughput computer vision-based methods [Luo et al., 2023, Kempelis et al., 2022, Shawon et al., 2024].

Regular RGB cameras for the online monitoring of plants are increasing in practical use. They are able to detect plant diseases that are analyzed from visible changes in plants, such as leaf blackening, colored spots on leaves, or altered plant morphology [Laura et al., 2024, Hernández et al., 2024, Polly and Devi, 2024]. Although RGB imaging offers valuable information for plant monitoring, it is limited to detecting and classifying diseases based on visible symptoms. This restriction reduces its effectiveness for growth modeling, and its accuracy often falls short of the requirements for analyzing and optimizing plant growth in precision agriculture. In addition, RGB-captured information is often impaired by illumination variance.

Hyperspectral imaging (HSI) is an emerging technology [Bian et al., 2024] that captures extensive structural, biochemical, and phenological data about plants, but mostly as a distanced remote sensing for urban study [Zhu et al., 2024, Huang et al., 2021, Hu et al., 2021] and crop field monitoring [Reddy et al., 2024, Lu et al., 2024]. Compared to RGB cameras, a hyperspectral camera collects light reflected from or transmitted through a plant. Since the light is split into many narrow wavelength bands using devices like prisms, diffraction gratings, or specialized filters, hyperspectral cameras can capture detailed spectral information at a high spectral resolution and across a wide range of wavelengths. Each pixel in a hyperspectral image

contains a spectrum that can be used to identify the composition and condition of the plant. Subtle variations in plant properties, physiological changes in plants, and biochemical properties can be identified and quantified to provide a more accurate classification of plant conditions, disease types, and maturity stages. This non-invasive and non-destructive method makes it ideal for monitoring living plants, e.g., detecting plant stress, nutrient levels, etc., and measuring plant traits such as chlorophyll content, water levels, or pigment distribution [Sarić et al., 2022]. Recently, hyperspectral imaging began to be utilized to provide rich information to detect diseases and growth status, as it mostly serves as a distanced remote sensing [Zhao et al., 2023b, Chun et al., 2024, Bao et al., 2024, Xu et al., 2023]. However, the technology’s higher cost, complexity, and computational demands compared to RGB systems may pose challenges for widespread adoption. In summary, extracting meaningful information from hyperspectral images remains a grand challenge. Traditional HSI classification methods [Thenkabail et al., 2013, Zhang et al., 2023b] often rely on pure spectral features or vegetation indices as an input, failing to make complete use of all available information.

Recently, deep neural network-based methods have started to be used for hyperspectral image processing [Dong et al., 2021, Lu et al., 2021]. Due to the multi-dimensionality and complexity of the data, obtaining effective feature representations across both the spatial and spectral domain remains an unsolved issue. Convolutional neural network-based methods [Kutluk et al., 2021, Slavkovikj et al., 2015] offer a simple and straightforward technique for plant growth modeling, but its inherited local receptive fields prevent the network from obtaining global relationships between the spatial, spectral, and temporal domains. As a result, the method’s own potential to utilize the full capabilities of hyperspectral imaging is limited. Instead, transformers have stronger parallelism and the ability to handle long-range dependencies compared to the Convolutional neural network. Because of this, transformer-based methods have reached adequate performance in a variety of applications in hyperspectral image analysis, such as super-resolution [Zhang et al., 2023a], classification [Scheibenreif et al., 2023, Zhao et al., 2023a], and denoising [Li et al., 2023]. However, due to the higher dimensionality, larger sizes, and more complicated nature of time-series hyperspectral images, the application of transformers into this category of HSI images has yet to be explored. Transformer-based time-series hyperspectral image analysis has the potential to pave new roads for advances in fundamental computational science as well as new applications, such as an accurate modeling of plant growth patterns under controlled environments.

In this project, I design a novel multimodal spatial-spectral-temporal transformer network to model plant growth with hyperspectral imaging. A hyperspectral camera is employed for the close-up sensing of plant leaves, and the network learns insight information about plant physiological conditions and developmental progression through time-series hyperspectral images. Furthermore, it captures the interactions between plants and environmental factors such as the light intensity and spectra, paving the way for precision control in vertical farming systems.

My contributions are summarized as follows:

- To integrate spatial context into the spectral sequence of each pixel, my spatial-spectral transformer divides the input image into 3D hyperspectral patches and utilizes a novel 3D aware positional encoding and the self-attention mechanism to capture global relations among these patches, leading to an effective feature representation of spatial and spectral information from the hyperspectral images.
- A temporal transformer architecture is proposed to model spatial-spectral representations at different time points and their long-range dependencies. The extracted accurate cross-dimensional features well capture the dynamics of the time-series hyperspectral images.
- The novel multimodal cross-attention mechanism models the interaction between environmental features and spatial-spectral-temporal representations from plant hyperspectral images, resulting in an accurate plant growth model.

- Extensive experiments demonstrate the performance of the multimodal spatial-spectral-temporal network. Evident by these experiments, an optimal strategy to control environment conditions for plant growth is presented.

2 Related Work

In this section, computer vision-based sensing and deep neural network-based AI methods for analyzing plant growth are reviewed.

2.1 Computer Vision-based Sensing of Plant's Status

Traditional sensing technologies for temperature, relative humidity, potential of hydrogen, etc. have been studied for a long time and progressed rapidly nowadays. In this work, I mainly focus on computer vision-based sensing technologies.

2.1.1 RGB Imaging-based Sensing

In the computer vision-based sensing field, regular RGB images are commonly used for diagnosing plant diseases that cause visible changes in the plants, such as root blackening, colored spots on leaves, or altered plant morphology. Laura et al. [Laura et al., 2024] proposed an aerial remote sensing system based on unmanned aerial vehicles (UAVs) and the YOLO v5x algorithm [Bolya et al., 2019]. This aerial remote sensing system uses an automatic UAV to capture high-definition RGB images to build a dataset of broccoli diseases/health conditions. The model trained with this dataset performs well in the detection of pathogens in broccoli. Hernandez et al. [Hernández et al., 2024] proposed an automated method based on interpretable deep learning for the detection and localization of plant disease symptoms under field conditions with a Canon EOS 5D Mark IV RGB camera. Combined with the sliding window method and interpretability analysis techniques, it achieved efficient and accurate disease detection and localization. This method has good generalization ability and is applicable to different grape varieties and field conditions. Polly et al. [Polly and Devi, 2024] proposed a plant leaf damage detection system that combines RGB image-level and pixel-level classification models. The CNN model is used for disease detection at the image level, while the U-Net model is applied for damage detection at the pixel level. This system outperforms existing methods in terms of disease recognition accuracy and damage detection precision on the Plant Village Dataset. Sajitha et al. [Sajitha et al., 2024] proposed a deep learning method based on a hybrid optimal attention capsule network for the automatic detection of diseases in pomegranate fruits. Through improved RGB image filtering and multi-feature extraction, the model combines the dynamic routing algorithm and the Aquila optimization algorithm to achieve a good performance in classification accuracy, precision, and recall rate. This method not only improved the accuracy of disease detection but also reduced false positives and false negatives. As a result, it has the potential to provide effective support for pomegranate cultivation.

Even though RGB imaging provides ample information for monitoring plants, they are only capable of detecting and classifying diseases based on appearance in the visible range of wavelength, which limits its scope in plant sensing and monitoring. Additionally, its accuracy is not sufficient enough to be applied into precision agriculture.

2.1.2 Hyperspectral Imaging-based Sensing

Most recently, hyperspectral imaging (HSI), a rising technology, has been used to provide rich information towards the detection of diseases and growth status. Furthermore, it is primarily utilized as a distanced remote sensing. Zhao et al. [Zhao et al., 2023b] used near-infrared HSI with random forests (RF), partial

least squares (PLS), and recurrent neural networks (RNN) to predict firmness, soluble solids, lycopene, and titratable acidity and classify the maturity stages of processing tomatoes. The RNN used showed good performance in quality prediction and maturity classification, which provides new insights in detecting the intrinsic quality and maturity of other agricultural products. Chun et al. [Chun et al., 2024] developed a method for the early detection of strawberry gray mold infection by utilizing hyperspectral fluorescence imaging and deep learning algorithms. Through the application of data augmentation techniques and spectral preprocessing techniques, this method improved the classification performance of the PLS-DA and 1D-CNN models. In particular, the ResNet-50 model demonstrated the highest classification accuracy when using mean normalization preprocessing and data augmentation techniques. Bao et al. [Bao et al., 2024] proposed a deep learning-based model architecture for detecting sugarcane diseases. Specifically, they used a modified ResNet34 model and combined it with a spectral-spatial attention mechanism to enhance their model’s performance. The experimental results showed that the model can achieve a relatively high detection accuracy before the disease symptoms even appeared. Xu et al. [Xu et al., 2023] demonstrated the effectiveness of corn seed disease recognition when Hyperspectral Imaging (HSI) and deep learning are used. The fusion of spectral and image features through the CNN improves the model’s recognition performance. The CNN-RB model further enhances the classification accuracy by strengthening its feature extraction capability. Information fusion strategies, especially advanced fusion strategies, can effectively compensate for the deficiencies of a single classifier and improve the overall classification performance. This method has great application potential in high-throughput phenotyping recognition systems. Ryckewaert et al. [Ryckewaert et al., 2023a] employed hyperspectral imaging to analyze grapevine leaves. The dataset contains hyperspectral data of grape leaves of seven distinct varieties. The samples contain both healthy leaves and those showing foliar symptoms of various grapevine diseases.

Compared to RGB imaging-based approaches, HSI imaging provides unique advantages for plant sensing and monitoring, particularly because of its ability to capture detailed spectral information at a high spectral resolution and across a wide range of wavelengths. This enables HSI to detect subtle variations in plant properties which are seemingly invisible when using RGB imaging. HSI can identify physiological changes in plants, such as stress or disease, even before visible symptoms appear. Additionally, biochemical properties like chlorophyll, flavonol, nitrogen, and water content, which facilitate the online monitoring of plant health and nutrient status, can be estimated using HSI. Conversely, RGB imaging lacks the spectral resolution to directly assess these properties. By leveraging spectral features, HSI is able to provide a more accurate classification of plant conditions, disease types, and maturity stages. However, HSI images are typically extremely large in size, requiring complex and computationally expensive methods as a result.

2.2 AI Techniques for Analyzing Plant Growth under Environmental Conditions

With the rapid development of AI, many types of deep neural network methods have been applied to plant monitoring.

2.2.1 Convolutional Neural Network

Because of the simplicity and strong capabilities of CNNs, many researchers have explored their application in the agricultural domain. Verma et al. proposed a deep learning model named BioNet [Verma et al., 2024]. This model utilizes a 1D Convolutional Neural Network (CNN) to extract spatial features from spectral data. It aims to optimize indoor agricultural practices by predicting plant growth under different LED light treatments in controlled agricultural environments. However, the use of only one external factor (i.e., light treatments) does not provide an accurate modeling of the interaction between the plant and its surrounding environment. Therefore, it leads to a suboptimal solution to address biomass prediction. Nejad et al. introduced the Capsule Attention Conv-LSTM Network (CACN) for soybean yield prediction [Ne-

jad et al., 2024]. By combining ConvLSTM and the attention mechanism, the CACN model can capture spatio-temporal dependencies and three-dimensional features in multispectral images. Their experimental results show that the CACN model is superior when compared to existing deep learning models in terms of prediction accuracy and robustness. This model performs well in extracting and representing features of different regions as well as capturing the complex relationships among different variables. However, the model does not take into account factors such as pest infestation, light intensity, and soil quality, which may have a significant impact on crop yield. Mohan et al. proposed a rice yield prediction model based on the Temporal Convolutional Network (TCN), which makes predictions through multispectral satellite data and climate parameters [Mohan et al., 2023]. The use of dilated convolutional layers effectively expands the receptive field and reduces the time and parameter complexity of the model. The Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and Two-Band Enhanced Vegetation Index (EVI2) are calculated using MODIS data. The study points out that the performance of the model is very sensitive to climate and vegetation values, which indicates that the model may not perform ideally under certain extreme conditions. Khan et al., 2024 [Khan et al., 2024] researched the possibility of using MODIS-derived Gross Primary Productivity (GPP) data and deep transfer learning to predict the yields of corn and soybeans in the United States Midwest. The results show that GPP data and deep transfer learning can effectively predict county-level crop yields, and the 1D-CNN model performs the best among them. The MODIS data used has a spatial resolution of 500 meters and may contain multiple crop types, resulting in errors in the final GPP estimation and affecting the accuracy of the yield prediction.

Even though CNN-based methods offer simple, straightforward, and plausible solutions for plant growth modeling, its inherited local receptive fields prevent it from obtaining global relationships in the spatial, spectral, and temporal domains, limiting its potential and accuracy.

2.2.2 Transformer

Many researchers have explored applying the Transformer to plant analysis tasks. Bi et al. proposed a transformer-based method to predict soybean yield through time-series RGB images and seed information [Bi et al., 2023]. Each RGB image in the time series is segmented into a plant part and a soil part, which are respectively processed by two Vision Transformers (ViT). The processed results are multiplied and combined to obtain a fused feature. The fused feature, together with the seed information, is then used as the input of a Fully Convolutional Network (FCN) to analyze the yield for crop fields. Onoufriou et al. [Onoufriou et al., 2023] developed a multi-timeline transformer network architecture, named Premonition Net, for strawberry yield prediction. The model consists of three different transformer encoders and decoders. These three different decoders are respectively targeted at past data, present data, and predicted future data, and the features extracted from these three aspects are then merged through a dense layer. In this work, fixed positional encodings are used to provide sequential information of the input data. Qiao et al. [Qiao et al., 2023] proposed a new Knowledge-guided Spatial-Temporal Attention Graph Network (KSTAGE) and a new KTMA algorithm, which automatically generates temporal attention weights under the guidance of prior knowledge through the self-attention mechanism. This is used to study how to accurately and timely predict crop yields, especially considering the nonlinear and dynamic spatio-temporal correlations during the plant growth process. The proposed model has demonstrated good performance in predicting winter wheat and corn in China and soybeans in the United States. Guo et al. proposed a deep learning framework based on the Transformer, called SSA-LSTM-Transformer (SLTF) [Guo et al., 2024]. The SLTF model is designed with a seven-layer structure, including an input layer, a Transformer encoding layer, an LSTM layer, a Transformer decoding layer, a dense layer, two dropout layers, and an output layer. The model is used to solve problems of nonlinear parameter optimization and long-term dependence of time-series data in winter wheat yield estimation. The predicted data indices include the Vegetation Temperature Condition Index, the Leaf Area Index, and the Fraction of Photosynthetically Active Radiation. Lin et al.

constructed a Multimodal Spatio-Temporal Vision Transformer (MMST-ViT) [Lin et al., 2023] for solving the problem of crop yield prediction at the regional county level in the United States. This transformer utilizes satellite images and short-term meteorological data to capture the impact of weather changes during the growing season on crop growth. The experiment used three types of data: crop data from the United States Department of Agriculture, High-Resolution Rapid Refresh (HRRR) meteorological data, and remote sensing data from the Sentinel-2 satellite. The results of the comparative experiments prove that MMST-ViT is superior to ConvLSTM, CNN-RNN, and GNN-RNN in all three performance indicators.

Since transformers have stronger parallelism and the ability to handle long-range dependencies as compared to CNN, RNN, and LSTM, they have reached great performance in a variety of applications in plant analysis. However, due to the higher dimensionality, large sizes, and complex nature of hyperspectral images, the application of transformers to time-series hyperspectral images has not been done. In this work, I will be the first to design a multimodal spatial-spectral-temporal transformer for time-series hyperspectral image analysis, aiming to accurately model plant status and growth patterns under controlled environments.

3 Novel Spatial-Spectral-Temporal Transformer for Multimodal Modeling of Time-series Hyperspectral Images

The modeling of environmental factors and the plants’ response to these factors is vitally important for establishing a plant growth model. In this study, I mainly investigate the different light treatments and the corresponding subtle variations in plant properties identified by advanced hyperspectral imaging. My proposed spatial-spectral-temporal transformer learns the knowledge from hyperspectral images and light treatments at different time series and models their interactions and relationships to generate an accurate growth modeling and final plant biomass prediction. In this work, I first propose to use hyperspectral reflectance imaging (HRI) to capture the plant’s characteristic properties (Section 3.1), then design a spatial-spectral embedding to capture fine-scale spatial and spectral feature representations (Section 3.2), and finally invent a multimodal spatial-spectral-temporal transformer for time-series modeling of these embedded spatial-spectral representations and their long-range dependencies to facilitate accurate analysis of plant properties (Section 3.3).

3.1 Hyperspectral Imaging and Reflectance Computation

As compared to RGB imaging, which is limited to only red, green and blue, HSI contains hundreds of spectral bands. This allows richer spectral information and subtle variations in properties which are invisible to RGB imaging to be captured and detected. For example, HSI can detect plant physiological changes, such as stress, using certain wavelengths associated with plant health. HSI can also estimate biochemical properties, including chlorophyll, flavonol, nitrogen, and water content, to infer plant health and nutrient status. On the contrary, RGB imaging lacks the spectral resolution to detect these properties, as it only detects color and morphological changes. Therefore, using specific spectral features, HSI facilitates a more accurate analysis of plant conditions for growth modeling.

Calculating reflectance in HSI images involves considering incident light. Therefore, a material capable of perfectly reflecting incident light must be utilized. By incorporating this material in the imaging process, reflectance can be calculated to build the plant field dataset. In the images, the intensity of the white areas corresponds to the incident light intensity. The intensity of the reflected light, $I(\lambda)$, was measured for each sample image. The dark current, $I_d(\lambda)$, (the signal recorded without illumination), was measured for each acquisition and subtracted from the data. Additionally, the intensity of light reflected by a certified standard reference provided by Specim IQ [Garcia et al., 2021], $I_0(\lambda)$, was recorded to standardize the spectra and mitigate nonlinearities caused by the components of the device. Using these measurements, a hyperspectral



Figure 1: Hyperspectral reflectance image example from my capture of a Radish sample. The spectrum profiles for a pixel of plant leaf (indicated by the green color dot), base (red color), background board (brown color), and soil (red color) are visualized.

reflectance image (HRI), $R(\lambda)$, is computed for each sample as in [Ryckewaert et al., 2023b] using the following formula:

$$R(\lambda) = \frac{I(\lambda) - I_d(\lambda)}{I_0(\lambda) - I_d(\lambda)} \quad (1)$$

where λ is the wavelength, $I(\lambda)$ and $I_0(\lambda)$ are the images of the reflected light intensity from the sample and reference respectively, and $I_d(\lambda)$ is the image in dark, which records the sensor signal in complete darkness (no light reaching the sensor). The dark current signal is subtracted from the raw hyperspectral data to remove unwanted noise from the thermal excitation of electrons within the sensor and improve the quality of the final reflectance data. Figure 1 shows the hyperspectral reflectance image of a plant, where the spectrum profiles for a pixel of plant leaf (indicated by the green color dot), base (red color), background board (brown color), and soil (red color) are visualized.

The computed hyperspectral reflectance images eliminate noise from external factors and standardize spectral information, which makes it easier to compare across samples and environments. Unlike traditional RGB images, hyperspectral reflectance images capture hundreds of narrow, contiguous spectral bands for each pixel, which provide ample quantitative information about the plant’s interaction with light across the spectrum, such as its absorption and scattering properties. This detailed spectrum allows for the identification of subtle variations in plant properties. Because it combines spatial information (image features) with spectral information (reflectance values), simultaneous analysis of the plant’s spatial and spectral properties can be facilitated. This is particularly useful for plant monitoring, where both spatial distribution and spectral signatures have significant impacts.

3.2 Spatial-Spectral Vision Transformer for Hyperspectral Image Embedding

To integrate spatial context into the spectral sequence of each pixel, my spatial-spectral model divides the input HRI image, I , into 3D patches. This approach is similar to the spatial patching in the original Vision Transformer (ViT) [Dosovitskiy et al., 2021]. However, it does not aggregate and summarize all spectral bands at each location as the previous work does [Yang et al., 2021, Hua et al., 2024]. Furthermore, it is able to preserve the hyperspectral 3D cube. Since Vision Transformer uses self-attention mechanisms to capture global dependencies and relationships between the divided hyperspectral patches directly across both spatial

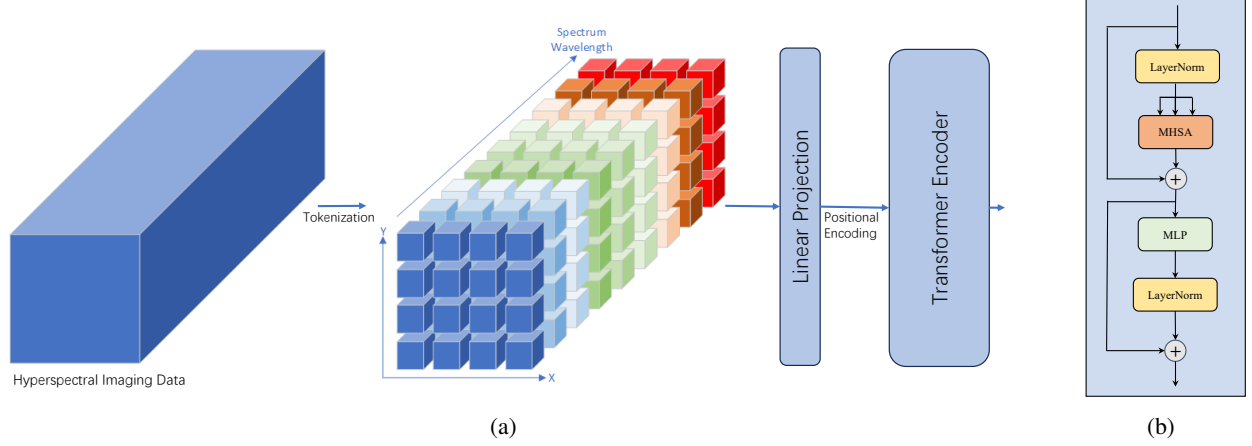


Figure 2: (a) My proposed spatial-spectral vision transformer, which encodes hyperspectral images across spatial and spectral dimensions and uses the attention mechanism to capture both spectral and spatial relationships within the image. (b) Illustration of multi-head self-attention (MHSA) mechanism within a transformer encoder.

and spectral dimensions, the model can effectively capture both spectral and spatial relationships within the image using attention mechanisms.

More specifically, as shown in Figure 2(a), hyperspectral reflectance imaging data, I , consisting of the two spatial dimensions, i.e., x and y , and the spectral dimension, i.e., λ , is divided into 3D patches. The hyperspectral imaging data is of $H \times W \times S$, with $H \times W$ indicating its spatial resolution and S as its spectral resolution. Assume the used patch size is of $P \times P \times C$. Each of the divided N patches ($N = (\frac{H}{P} \times \frac{W}{P} \times \frac{S}{C})$) are flattened into a one-dimensional vector by concatenating the spectral values. The flattened vector has $P \times P \times C$ elements. The flattened vector is then passed through a linear projection layer (i.e., fully connected layer), projecting to a fixed embedding dimension D . The learned weights in the linear projection layer implicitly capture inter-spatial and inter-spectral relationships simultaneously during training. The dimensionality-reduced output combines the low-dimensional embeddings from both spaces. As a result, it maintains the spatial structure and spectral channel information of the hyperspectral imaging patches, allowing the following transformer encoder to learn richer global representations of both spatial and spectral information. Mathematically,

$$h = Wp + b, \quad (2)$$

where h is the embedding feature, W and b are learned weight and bias parameters, respectively, and p is the flattened patch vector.

Since hyperspectral reflectance images contain spatial and spectral information that both need to be preserved during processing, these tokenized features are positionally encoded so that spatial-spectral context is added to the sequence of spectral-spatial patches, which allows the transformer to interpret spatial and spectral relationships more effectively. A novel 3D-aware positional encoding through a learnable multi-layer perceptron mapping (MLP) is proposed. It first computes normalized 3D coordinates for each patch as follows,

$$\begin{cases} P_{i_x}^{3D} = (P_{i_x}^{3D} - x_{min}) / (x_{max} - x_{min}) \\ P_{i_y}^{3D} = (P_{i_y}^{3D} - y_{min}) / (y_{max} - y_{min}) \\ P_{i_\lambda}^{3D} = (P_{i_\lambda}^{3D} - \lambda_{min}) / (\lambda_{max} - \lambda_{min}) \end{cases}$$

where $(P_{i_x}, P_{i_y}, P_{i_\lambda})$ is the 3D coordinates of the i -th patch., $x_{min}, y_{min}, \lambda_{min}$ and $x_{max}, y_{max}, \lambda_{max}$ represent the bounding box. Then the sine functions of these coordinates are mapped by the MLP to the

corresponding model dimension as follows,

$$SSPE_i = \text{MLP}(\text{Cat}(\text{sine}(P_{i_x}^{3D}), \text{sine}(P_{i_y}^{3D}), \text{sine}(P_{i_\lambda}^{3D}))) \quad (3)$$

where $\text{sine} = \sin(\text{pos}/10000^{2j/d_{\text{model}}})$, and $SSPE_i$ is the positional encoding for the spatial-spectral transformer.

After adding the 3D-aware positional encoding, they are input into the transformer encoder (see Figure 2(b)) to form latent representations with global dependency information. These latent representations obtain rich information from the tokens (including both spatial and spectral information) and their contextual relationships in the spatial-spectral space. Similar to Vision Transformer (ViT) [Dosovitskiy et al., 2021], my transformer encoder only uses the transformer encoder part in ViT (see Figure 2(b)). The self-attention mechanism within the transformer encoder extracts global features, which capture correlations between tokens to represent significant long-range contextual dependencies along both the spatial and spectral dimensions.

In this process, each token generates an attention score using its own query vector and the key vectors from other tokens, detecting its relative importance to other tokens. Using these scores, each token performs a weighted average of the value vectors of those tokens to produce its output vector. This approach essentially enables each token to integrate information from other tokens. As a result, global feature extraction is facilitated. Such capabilities are extremely important for detecting small changes in the spatial and spectral dimensions of hyperspectral images. The self-attention mechanism [Vaswani et al., 2017] is mathematically defined as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V, \quad (4)$$

where Q , K , and V are the query, key, and value projections. The dot product of Q and K computes the attention score for each vector. The term, d_k is the dimensionality of the key vector K and is used to scale scores by dividing by $\sqrt{d_k}$. This improves numerical stability during gradient computation. The scores are then normalized using the softmax function, which highlights the relative importance of each vector. Finally, the normalized scores are multiplied with the value vectors V to focus on relevant information.

To enhance the model's global feature extraction and generalization capabilities, the transformer encoder employs the multi-head self-attention mechanism (MHSA). The self-attention mechanism is divided into multiple parallel sub-modules by MHSA. Therefore, it allows the model to focus on different features and relationships simultaneously. This approach enhances the model's representation power and overall performance. In MHSA, the input is split into M heads ($\text{head}_1, \dots, \text{head}_M$), with each head independently computing self-attention. The results from all heads are then concatenated to form the final output. The formulas for MHSA are as follows:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad (5)$$

$$\text{MHSA}(Q, K, V) = \text{Concat}((\text{head}_1, \dots, \text{head}_i, \dots, \text{head}_M)W^0), \quad (6)$$

where W_i^Q , W_i^K , W_i^V ($i = 1, 2, \dots, M$), and W^0 are learnable weight matrices and are correspondent to each head and the final output, respectively. MHSA facilitates simultaneous spatial and spectral processing by adaptively allocating attention weights to both spatial and spectral features, ensuring that relevant information is prioritized according to the objective task. Additionally, each head in MHSA can focus on different spatial locations or spectral bands independently, allowing the model to capture fine-scale spatial and spectral relationships. In addition, the self-attention mechanism allows the encoder to consider long-range dependencies across both spatial and spectral dimensions, which is essential for hyperspectral images, as correlations between distant spectral bands and spatial locations can provide critical information for prediction tasks.

3.3 Novel Spatial-Spectral-Temporal Transformer for Multimodal Time-Series Modeling of Plant Growth

In order to model the time-series hyperspectral reflectance imaging of plant growth, a sequence of hyperspectral images at different time points, $\mathbf{I} = \{I_{t_0}, I_{t_1}, \dots, I_{t_n}\}$, are taken as the input. Here, the objective is to find a regressive function to compute the output biomass, Y , based on these input time-series hyperspectral images, $\mathbf{I}$, and their corresponding light treatment, $\mathbf{L}$.

$$Y = \phi(\mathbf{I}, \mathbf{L}). \quad (7)$$

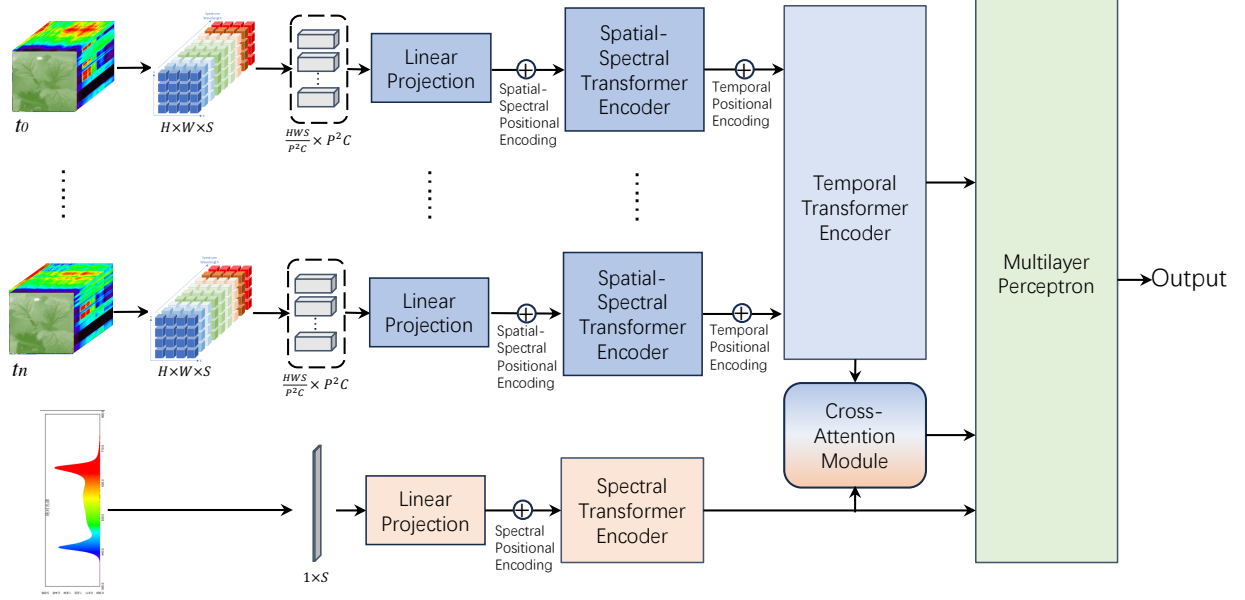


Figure 3: My proposed spatial-spectral-temporal transformer for early estimation of the biomass based on hyperspectral images and light treatment (i.e., light spectra and intensities) over a certain lifespan of the plant.

3.3.1 Spatial-Spectral-Temporal Transformer

To facilitate the effective modeling of yield-related features and their long-term dependencies in the spatial, spectral, and timing dimensions in time-series HRI data, my novel spatial-spectral-temporal transformer uses self-attention mechanisms to compute relationships between all data points in the sequence, regardless of distance, as shown in Figure 3. The HRI images are first handled by the spatial-spectral vision transformer encoder to obtain latent representations across the spatial and spectral domains as described in Section 3.2. Then, a temporal transformer encoder is subsequently used to further enhance their feature representations for the regression purpose. Similar to the spatial-spectral vision transformer as shown in Figure 2, the encoder consists of a temporal positional encoding layer and multiple stacked multi-head attention layers. Positional encoding, implemented with sine and cosine functions, incorporates sequential information into the time-series HRI features by adding the feature vector element into a positional encoding vector. Each multi-head attention layer calculates the attention coefficients between the features at different timestamps. Finally, an output layer maps the results from the last multi-head attention layer into latent feature representations. The self-attention mechanism in the spatial-spectral-temporal transformer allows the model to focus on the most relevant patterns in the spatial-spectral domains and their time series.

3.3.2 Cross-Attention Exploration of Multimodal Relationship

In order to effectively model the interaction between HRI images and light treatments, I integrate a one-dimensional transformer encoder to process light treatment ranging from 350 nm to 800 nm. The light treatment is considered as a discretized vector, $\mathbf{L} = \{l_0, l_1, \dots, l_{450}\}$. The input is mapped through a linear projection and is then positionally encoded based on its spectral band position. Similar to other transformers, a spectral transformer encoder is employed to model the light spectrum representation and their long range dependencies. The spectral transformer's encoded representations of light treatment, along with the output from the transformer encoders (for time-series hyperspectral images), are fed to a multilayer perceptron for regression analysis.

In order to further explore the interactions and relationship between time-series HRI images and light spectra data, a cross-attention module is designed to combine the outputs of both encoders to learn interactions and explicitly compute relationships between time-series HRI representations and light spectral representations as shown in Figure 3. It creates a mechanism where one data modality (i.e., light spectral features) guides the feature extraction process for the other modality (i.e., spatial-spectral-temporal features of time-series HRI images). The model can therefore learn how light spectrum influences spatial-spectral-temporal information from time-series hyperspectral images and vice versa.

Suppose the spatial-spectral-temporal encoder produces latent embeddings, E_{HRI_T} , and the light spectral transformer encoder produces latent embeddings, E_{light} . My cross-attention module takes E_{light} to generate the query, Q_{light} , and use E_{HRI}^T to compute the key K_{HRI_T} and value V_{HRI_T} . The cross-attention mechanism is computed as follows,

$$\text{CrossAttention}(Q_{light}, K_{HRI_T}, V_{HRI_T}) = \text{softmax} \left(\frac{Q_{light} K_{HRI_T}^T}{\sqrt{d_k}} \right) \cdot V_{HRI_T}. \quad (8)$$

Different from the query, key, and value vectors in the self-attention mechanism, in Equation 4, the query Q_{light} here is computed from a different modality than the key, K_{HRI_T} , and value, V_{HRI_T} . Applying cross-attention to fuse these features enables the model to predict growth with a better understanding of how light impacts spatial-spectral-temporal patterns. Similar to the self-attention mechanism, the cross-attention mechanism can be applied with the standard multi-head attention approach. The input data of the mechanism is separated into multiple heads, performing cross-attention computations independently for each head and then concatenating the results. The resulting cross-attention latent representation, along with the spectral transformer's encoded representation of light treatment and the output from the transformer encoders for time-series hyperspectral images, are fed into a multilayer perceptron for further regression analysis.

3.3.3 Loss Function

The entire spatial-spectral-temporal transformer network allows for end-to-end training, reducing the need for preprocessing steps and enabling robust feature extraction directly from hyperspectral data. To supervise the learning of the spatial-spectral-temporal transformer network, mean squared error (MSE) is used to compare the biomass prediction to the ground truth. Mean Squared Error (MSE) loss calculates the mean of the squared differences between the predicted and true values. The equation for MSE is as follows,

$$\text{MSE}_{\text{biomass}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (9)$$

where, y is the ground truth and $\hat{y}$ is the predicted biomass. By minimizing mean square error, the prediction function, $Y = \phi(\mathbf{I}, \mathbf{L})$, can be established based on input HRI image sequences and their specified light treatment.

4 System Development and Experiments

In the experiments, my primary objective is to explore the relationship between light treatment and plant biomass and establish a growth model with collected data. To ensure the reliability of my experiments, I maintain other growth factors, including relative humidity, temperature, soil pH, and lighting duration, constant through all the growth batches. I developed a small indoor vertical farming system using commercial LED lights, which are able to generate light spectra with input coefficients.

4.1 Experiment Condition and Setup

Commercially available Radish (*Raphanus sativum* cv Champion) seeds were purchased for the experiments in this project. All seeds were germinated by the Jiffy seed starting tray kit under controlled environmental conditions with photosynthetic photon flux density (PPFD) of $150 \mu\text{mol}/\text{m}^2/\text{s}$, a photoperiod of 14 hours per day, an air temperature of $68 \pm 2^\circ\text{F}$, and a relative humidity of $70 \pm 5\%$. After sprouting, seedlings were transplanted into controlled environmental conditions under different spectral light treatment, including a photoperiod of 14 hours per day, an air temperature of $68 \pm 2^\circ\text{F}$, a relative humidity of $70 \pm 5\%$, and this was all conducted in a simulated indoor vertical farming system.

In the experiment, the light spectrum was generated and controlled by adjustable LEDs (provided by Paul Lee Co., Ltd., California) with white, red, blue, and IR lights, which all can be adjusted from a power level of 0% to 100% to generate the desired spectral light treatment conditions. Figure 4 illustrated PPFD of each color LED light at a power level of 100%.

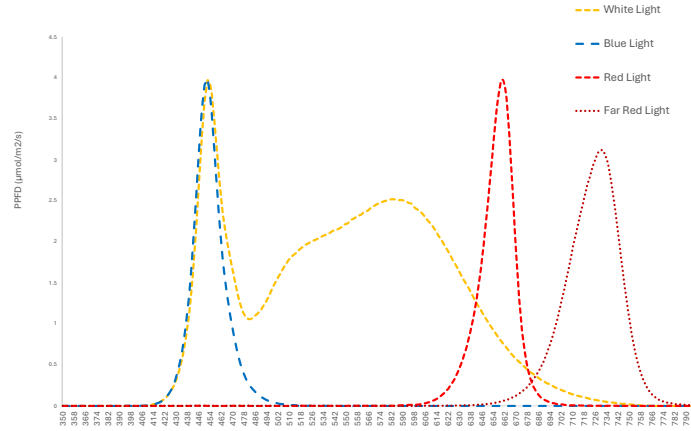


Figure 4: The adjustable LEDs consists of white, red, blue, IR lights. Photosynthetic Photon Flux Density (PPFD) of each light of the four lights at 100% power level was measured by a spectrometer (EVERFINE PLA-30).

Each batch was treated under different spectral light treatments with a PPFD ranging from $245 - 560 \mu\text{mol}/\text{m}^2/\text{s}$ and Red/Blue light PPFD ratio ranging from 0.6 – 4.0 as shown in Figure 5. A total of 96 different light treatments were employed. The LED lights used in these tests were installed horizontally at 70cm above the cultivation bench. LED Light photosynthetic photon flux density (PPFD) was validated using a Quantum PAR Full Spectrum (DP-355, DANOPLUS).

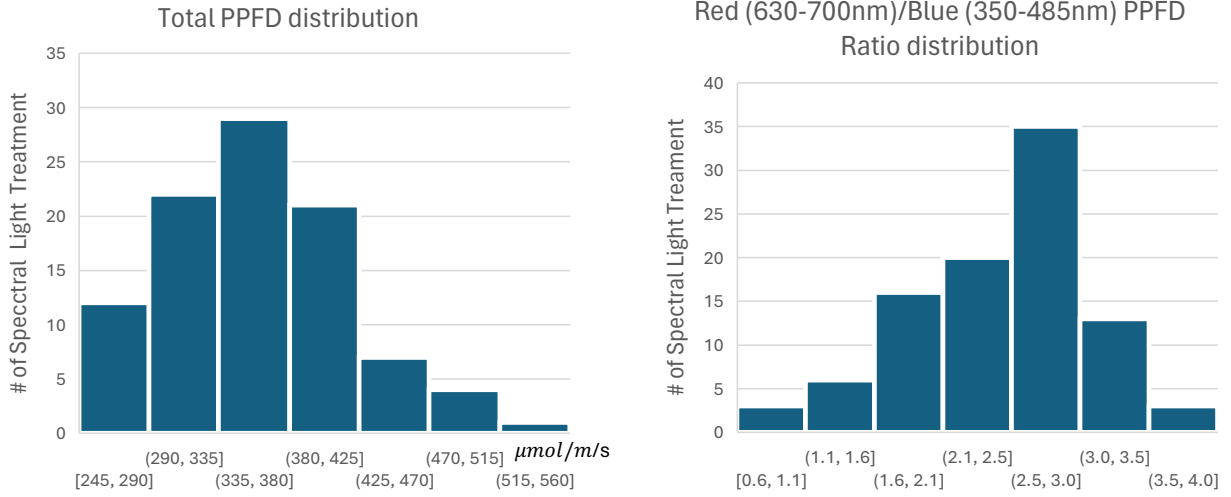


Figure 5: Each batch was treated by different combined total PPFD and Red (630 – 700nm)/Blue (350 – 485nm) PPFD Ratios generated by adjusting different LED Lights (W, R, IR, B) power levels.



Figure 6: (a) The simulated indoor farm with controllable LED lights; (b) The plant images were acquired using a hyperspectral camera (IQ, Specim, Finland). A constant angle of 85 degrees between the LED lamp axis and the hyperspectral camera axis was always maintained.

4.2 Hyperspectral Image and Growth Data Collection

Hyperspectral images were captured for each individual plant under controlled field conditions with a portable hyperspectral camera (SPECIM IQ with full spectrum 400–1000 nm, Finland) mounted on a tripod at a height of 55 cm. Hyperspectral images for each plant were taken on a daily basis within the first hour after the photo-period started. The used spectral resolution is of 7 nm. The ambient illumination was provided by a white light positioned above the samples. An acquisition angle of 85 degrees between the LED lamp axis and the hyperspectral camera axis is used, as seen in Figure 6, and the illumination was provided by a white LED light above the samples.

A total of 96 batches with 6 radish plants in each batch were studied. After a 12 day growing period after the transplanting, each plant was extracted out of the soil to measure the selected growth parameter,

i.e., full plant fresh biomass. The fresh plant’s weight was measured using a digital scale (0.01g).

5 example batches were illustrated in Figure 7 and Table 1 to demonstrate different spectral treatments and corresponding growth measurements collected. Some samples of the harvested radishes on Day 12 were shown in Figure 8(a) where the bottom shows their weights. Figure 8(b) is the weight distribution for all the samples in my data collection.

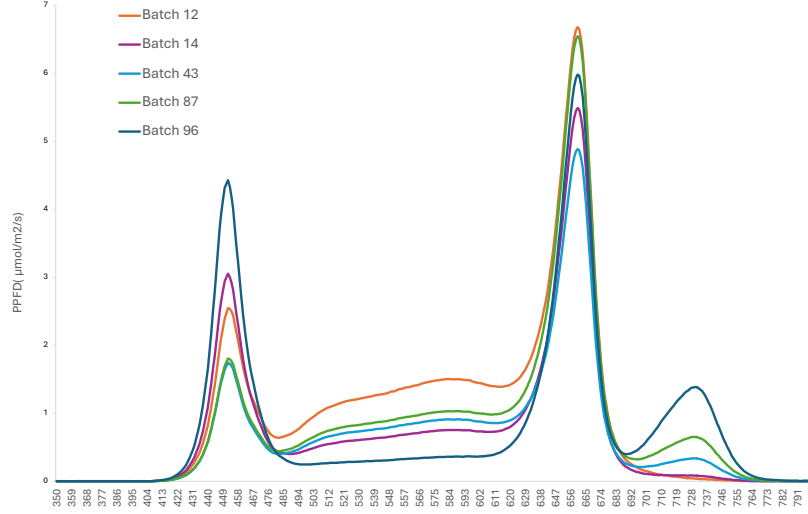


Figure 7: 5 sample light treatments with different PPFDs and PARs.

Batch #	12	14	43	87	96
PFD in 350-485nm	70	77	47	49	107
PFD in 486-629nm	184	95	112	128	50
PFD in 630-700nm	190	150	138	183	163
PFD in 700-800nm	3	4	14	27	55
Total PPFD	447	326	311	387	375
PFD% in 350-485nm	16%	24%	15%	13%	29%
PFD% in 486-629nm	41%	29%	36%	33%	13%
PFD% in 630-700nm	42%	46%	44%	47%	43%
PFD% in 700-800nm	1%	1%	5%	7%	15%
(630-700nm)/(350-485nm) PPFD Ratio	2.7	1.9	2.9	3.7	1.5
Mean Biomass (gram)	1.81	1.91	2.17	2.35	1.83

Table 1: 5 different spectral treatments as shown in Figure 7 and their corresponding growth measurements.

4.3 Algorithm Setup, Implementation and Training Details

The input of a hyperspectral reflectance image is of size $512 \times 512 \times 132$. The patch size used in the spatial-spectral transformer is set as $4 \times 4 \times 4$. For the temporal transformer, I used three different settings, resulting in three different biomass estimation models. *Model₄* uses *Day2* and *Day4* HRI images as the input, *Model₆* uses *Day2*, *Day4*, and *Day6* images, and *Model₈* uses *Day2*, *Day4*, *Day6*, and *Day8* images as the input. The temporal transformer encoder, light spectral transformer, and cross-attention module all have



Figure 8: (a) Some samples of the harvested radishes on Day 12; (b) The weight distribution of all the samples in my data collection.

outputs of dimensions 64, which are fed together into the MLP with three layers (192, 96, 1 neurons at each layer) to produce the final prediction. Hence, my development produces three biomass prediction models upon three different time-series HRI inputs.

I used PyTorch to develop my deep learning network, which is trained and tested on a single NVIDIA GeForce RTX 3090 GPU with 24GB GDDR6X. The Adam optimization method [Kingma and Ba, 2014] with weight decay of 0.00001 is used for model parameter optimization. The initial learning rate is 0.0001 and the learning rate was gradually reduced over 100 epochs by the cosine annealing algorithm.

The training set of plants' hyperspectral images and growth parameters collected were used in AI model training. Among the 96 batches with 6 radish plants in each batch, I collected 6912 HSI images from 576 radish samples to form the database. In addition, I added 24 background hyperspectral images (containing the capture of the soil, cup, ground, etc. without plants) with final biomasses set to 0. Each data sample contains a sequence of HRI images, the corresponding light treatment used, and the final biomasses. The random sample sets of 480, 60, and 60 units are used for training, validation, and testing, respectively.

4.4 Quantitative Metrics for Performance Evaluation

Similar to another work [Alhnaity et al., 2019], four metrics: Root mean square error (RMSE), R squared value (R^2), mean absolute error (MAE), and mean absolute percentage error (MAPE), are used to evaluate the performance of my method. The computation of RMSE, R, and MAE is as follows,

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{y - \hat{y}}{y} \right)^2}, \quad (10)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y - \hat{y})^2}{\sum_{i=1}^N (y - \bar{y})^2}, \quad (11)$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N \|y - \hat{y}\|, \quad (12)$$

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \frac{\|y - \hat{y}\|}{\|y\|}, \quad (13)$$

where y is the ground truth and $\hat{y}$ is the predicted biomass.

4.5 Evaluation and Comparison of the Multimodal Spatial-Spectral-Temporal Transformer

The multimodal spatial-spectral-temporal transformer was evaluated on the testing data with 60 samples. As aforementioned, *Model₄* uses *Day2* and *Day4* HRI images as the input, *Model₆* uses *Day2*, *Day4*, and *Day6* images, and *Model₈* uses *Day2*, *Day4*, *Day6*, and *Day8* images as the input.

Model	<i>Model₄</i>	<i>Model₆</i>	<i>Model₈</i>	<i>SVR₈</i>
RMSE ↓	0.219	0.116	0.061	0.205
R ² ↑	0.795	0.890	0.919	0.759
MAE ↓	0.192	0.098	0.056	0.258
MAPE ↓	0.107	0.056	0.030	0.151

Table 2: Evaluation of the multimodal spatial-spectral-temporal transformer network on the testing dataset. *Model₄* uses *Day2* and *Day4* HRI images as the input, *Model₆* uses *Day2*, *Day4*, and *Day6* images, and *Model₈* uses *Day2*, *Day4*, *Day6*, and *Day8* images.

Based on the same training, validation, and testing data split, the multimodal spatial-spectral-temporal transformer was compared to a traditional support vector regression (SVR) method. For the SVR method, the NDVI [Thenkabail et al., 2000] index serves as a measure for hyperspectral reflectance images. The NDVIs from *Day2*, *Day4*, *Day6*, and *Day8* images and light treatment vectors were used to produce a prediction with the SVR method. As seen in Table 2, my model consistently outperforms the SVR method.

4.6 Model Analysis and Ablation Study

I conducted ablation study and model analysis for the designed network. The model analysis and ablation study were completed by removing one component each time, i.e., without the cross-attention module, spectral transformer, or spatial-spectral-temporal transformer. The training, validation, and testing were conducted with the same dataset splits, and the performance was recorded in Table 3.

Model	<i>Model₈</i>	w/o cross-attention	w/o spectral t.	w/o spatial-spectral-temporal t.
RMSE ↓	0.061	0.073	0.077	0.133
R ² ↑	0.919	0.829	0.815	0.782
MAE ↓	0.056	0.080	0.128	0.209
MAPE ↓	0.030	0.047	0.070	0.115

Table 3: Model analysis and ablation study of my Spatial-Spectral-Temporal transformer network. The model analysis and ablation study were done by removing one component each time, i.e., without the cross-attention module, spectral transformer, or spatial-spectral-temporal transformer.

4.7 Improved Light Treatment Strategy

Based on the range of PPFD and the Red/Blue light ratio, I continuously randomize the ratio and PPFD value to generate new light treatment, which is then input into the model without the spatial-spectral-temporal transformer for predicting biomass. When a higher yield is found, I output the corresponding light treatment as a fine-tuned and improved lighting strategy. In this experiment, I found a better light treatment strategy compared to those used in the previous dataset. The light treatment is illustrated in Figure 9. In practice, this can be conducted at different time points during the plant growth period to redesign the improved light treatment, allowing for a dynamic treatment strategy.

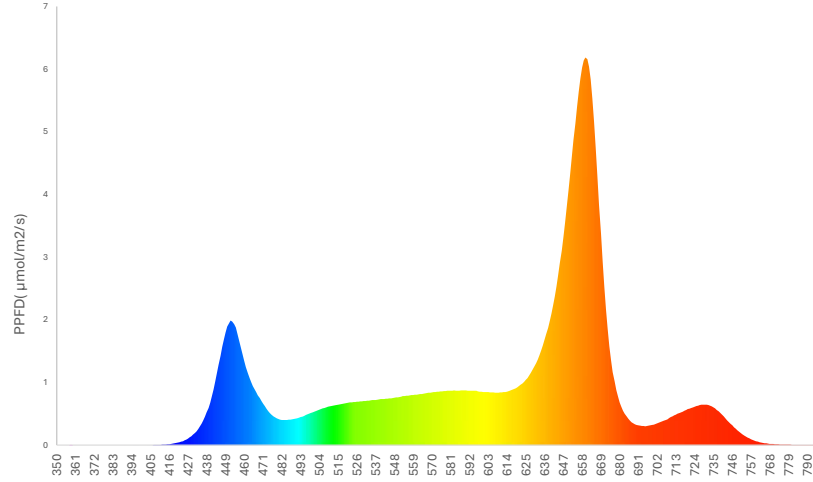


Figure 9: (a) Some samples of the harvested radishes on Day 12; (b) The weight distribution of all the samples in my data collection.

5 Conclusion

In this work, I have presented a novel multimodal spatial-spectral-temporal transformer to integrate spatial, spectral, and temporal features of a plant’s early lifespan to yield a statistical model representing the plant’s growth patterns in response to environmental factors. This model is trained using the hyperspectral images of the plant, the data of the environmental conditions (light spectra and intensities), and the plant’s final biomass to accurately model the interaction between environmental features and spatial-spectral-temporal representations from plant hyperspectral images. As a result, an accurate plant growth model is developed. Extensive experiments indicate the effectiveness of the proposed network. The MSST Transformer-based time-series hyperspectral image analysis paves the road for advances in both fundamental computational science as well as new applications, such as precision agriculture, urban studies, etc.

In the future, I plan to investigate generative AI methods for automatic environmental condition control. Given that hyperspectral imaging shows the current plant’s status, a diffusion network can be designed to generate optimal environmental conditions for plant growth, leading to dynamic and adaptive environmental control. It can also be expanded to other plants and environmental factors to allow precise and dynamic environmental control for comprehensive smart farming and extreme condition farming.

6 References

- [Alhnaity et al., 2019] Alhnaity, B., Pearson, S., Leontidis, G., and Kollias, S. (2019). Using deep learning to predict plant growth and yield in greenhouse environments. In *International Symposium on Advanced Technologies and Management for Innovative Greenhouses*, pages 425–432.
- [Bao et al., 2024] Bao, D., Zhou, J., Bhuiyan, S. A., Adhikari, P., Tuxworth, G., Ford, R., and Gao, Y. (2024). Early detection of sugarcane smut and mosaic diseases via hyperspectral imaging and spectral-spatial attention deep neural networks. *Journal of Agriculture and Food Research*, 18:101369.

- [Bi et al., 2023] Bi, L., Wally, O., Hu, G., Tenuta, A. U., Kandel, Y. R., and Mueller, D. S. (2023). A transformer-based approach for early prediction of soybean yield using time-series images. *Frontiers in Plant Science*, 14:1173036.
- [Bian et al., 2024] Bian, L., Wang, Z., Zhang, Y., Li, L., Zhang, Y., Yang, C., Fang, W., Zhao, J., Zhu, C., Meng, Q., et al. (2024). A broadband hyperspectral image sensor with high spatio-temporal resolution. *Nature*, 635(8037):73–81.
- [Bolya et al., 2019] Bolya, D., Zhou, C., Xiao, F., and Lee, Y. J. (2019). YOLACT: real-time instance segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- [Chun et al., 2024] Chun, S.-W., Song, D.-J., Lee, K.-H., Kim, M.-J., Kim, M. S., Kim, K.-S., and Mo, C. (2024). Deep learning algorithm development for early detection of botrytis cinerea infected strawberry fruit using hyperspectral fluorescence imaging. *Postharvest Biology and Technology*, 214:112918.
- [Demotes-Mainard et al., 2016] Demotes-Mainard, S., Péron, T., Corot, A., Bertheloot, J., Le Gourrierc, J., Pelleschi-Travier, S., Crespel, L., Morel, P., Huché-Thélier, L., Boumaza, R., et al. (2016). Plant responses to red and far-red lights, applications in horticulture. *Environmental and experimental Botany*, 121:4–21.
- [Dong et al., 2021] Dong, W., Zhou, C., Wu, F., Wu, J., Shi, G., and Li, X. (2021). Model-guided deep hyperspectral image super-resolution. *IEEE Transactions on Image Processing*, 30:5754–5768.
- [Dosovitskiy et al., 2021] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*.
- [Garcia et al., 2021] Garcia, S. M., Ryckewaert, M., Abdelghafour, F., Metz, M., Moura, D., Feilhes, C., Prezman, F., and Bendoula, R. (2021). Combination of multivariate curve resolution with factorial discriminant analysis for the detection of grapevine diseases using hyperspectral imaging. a case study: Flavescence dorée. *Analyst*, 146(24):7730–7739.
- [Guo et al., 2024] Guo, F., Wang, P., Tansey, K., Zhang, Y., Li, M., Liu, J., and Zhang, S. (2024). A novel transformer-based neural network under model interpretability for improving wheat yield estimation using remotely sensed multi-variables. *Computers and Electronics in Agriculture*, 223:109111.
- [Hernández et al., 2024] Hernández, I., Gutiérrez, S., Barrio, I., Íñiguez, R., and Tardaguila, J. (2024). In-field disease symptom detection and localisation using explainable deep learning: Use case for downy mildew in grapevine. *Computers and Electronics in Agriculture*, 226:109478.
- [Hu et al., 2021] Hu, J.-F., Huang, T.-Z., Deng, L.-J., Jiang, T.-X., Vivone, G., and Chanussot, J. (2021). Hyperspectral image super-resolution via deep spatio-spectral attention convolutional neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 33(12):7251–7265.
- [Hua et al., 2024] Hua, M. J., Wu, J., and Zhong, Z. (2024). Multi-scale knowledge transfer vision transformer for 3d vessel shape segmentation. *Computers & Graphics*, 122:103976.
- [Huang et al., 2021] Huang, Y., Shen, Q., Fu, Y., and You, S. (2021). Weakly-supervised semantic segmentation in cityscape via hyperspectral image. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1117–1126.

- [Kaiser et al., 2024] Kaiser, E., Kusuma, P., Violet-Chabrand, S., Folta, K., Liu, Y., Poorter, H., Woning, N., Shrestha, S., Ciarreta, A., van Brenk, J., et al. (2024). Vertical farming goes dynamic: optimizing resource use efficiency, product quality, and energy costs. *Frontiers in Science*, 2:1411259.
- [Kempelis et al., 2022] Kempelis, A., Romanovs, A., and Patlins, A. (2022). Using computer vision and machine learning based methods for plant monitoring in agriculture: A systematic literature review. In *2022 63rd International Scientific Conference on Information Technology and Management Science of Riga Technical University (ITMS)*, pages 1–6. IEEE.
- [Khan et al., 2024] Khan, S. N., Li, D., and Maimaitijiang, M. (2024). Using gross primary production data and deep transfer learning for crop yield prediction in the us corn belt. *International Journal of Applied Earth Observation and Geoinformation*, 131:103965.
- [Kingma and Ba, 2014] Kingma, D. P. and Ba, J. (2014). Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*.
- [Kutluk et al., 2021] Kutluk, S., Kayabol, K., and Akan, A. (2021). A new cnn training approach with application to hyperspectral image classification. *Digital Signal Processing*, 113:103016.
- [Laura et al., 2024] Laura, D., Urrutia, E. P., Salazar, F., Ureña, J., Avalos, D. H., Célleri, L. A., Cazorla-Logroño, M., and Altamirano, S. (2024). Aerial remote sensing system to control pathogens and diseases in broccoli crops with the use of artificial vision. *Smart Agricultural Technology*, page 100739.
- [Li et al., 2023] Li, M., Liu, J., Fu, Y., Zhang, Y., and Dou, D. (2023). Spectral enhanced rectangle transformer for hyperspectral image denoising. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5805–5814.
- [Lin et al., 2023] Lin, F., Crawford, S., Guillot, K., Zhang, Y., Chen, Y., Yuan, X., Chen, L., Williams, S., Minvielle, R., Xiao, X., et al. (2023). Mmst-vit: Climate change-aware crop yield prediction via multi-modal spatial-temporal vision transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5774–5784.
- [Lu et al., 2021] Lu, X., Zhang, J., Yang, D., Xu, L., and Jia, F. (2021). Cascaded convolutional neural network-based hyperspectral image resolution enhancement via an auxiliary panchromatic image. *IEEE Transactions on Image Processing*, 30:6815–6828.
- [Lu et al., 2024] Lu, X., Zhang, J., Yang, R., Yang, Q., Chen, M., Xu, H., Wan, P., Guo, J., and Liu, F. (2024). Effective variance attention-enhanced diffusion model for crop field aerial image super resolution. *ISPRS Journal of Photogrammetry and Remote Sensing*, 218:50–68.
- [Luo et al., 2023] Luo, J., Li, B., and Leung, C. (2023). A survey of computer vision technologies in urban and controlled-environment agriculture. *ACM Computing Surveys*, 56(5):1–39.
- [Massa et al., 2008] Massa, G. D., Kim, H.-H., Wheeler, R. M., and Mitchell, C. A. (2008). Plant productivity in response to led lighting. *HortScience*, 43(7):1951–1956.
- [Meeker et al., 2021] Meeker, E. W., Magney, T. S., Bambach, N., Momayyezi, M., and McElrone, A. J. (2021). Modification of a gas exchange system to measure active and passive chlorophyll fluorescence simultaneously under field conditions. *AoB Plants*, 13(1):plaa066.
- [Mohan et al., 2023] Mohan, A., Venkatesan, M., Prabhavathy, P., and Jayakrishnan, A. (2023). Temporal convolutional network based rice crop yield prediction using multispectral satellite data. *Infrared Physics & Technology*, 135:104960.

- [Nejad et al., 2024] Nejad, S. M. M., Abbasi-Moghadam, D., Sharifi, A., and Tariq, A. (2024). Capsular attention conv-lstm network (cacn): A deep learning structure for crop yield estimation based on multi-spectral imagery. *European Journal of Agronomy*, 161:127369.
- [Onoufriou et al., 2023] Onoufriou, G., Hanheide, M., and Leontidis, G. (2023). Premonition net: a multi-timeline transformer network architecture towards strawberry tabletop yield forecasting. *Computers and Electronics in Agriculture*, 208:107784.
- [Polly and Devi, 2024] Polly, R. and Devi, E. A. (2024). Semantic segmentation for plant leaf disease classification and damage detection: A deep learning approach. *Smart Agricultural Technology*, 9:100526.
- [Qiao et al., 2023] Qiao, M., He, X., Cheng, X., Li, P., Zhao, Q., Zhao, C., and Tian, Z. (2023). Kstage: A knowledge-guided spatial-temporal attention graph learning network for crop yield prediction. *Information Sciences*, 619:19–37.
- [Reddy et al., 2024] Reddy, K. K., Daduvy, A., Mohana, R. M., Assiri, B., Shuaib, M., Alam, S., and Sheneamer, A. (2024). Enhancing precision agriculture and land cover classification: A self-attention 3d convolutional neural network approach for hyperspectral image analysis. *IEEE Access*.
- [Ryckewaert et al., 2023a] Ryckewaert, M., Hérán, D., Trani, J.-P., Mas-Garcia, S., Feilhes, C., Prezman, F., Serrano, E., and Bendoula, R. (2023a). Hyperspectral images of grapevine leaves including healthy leaves and leaves with biotic and abiotic symptoms. *Scientific Data*, 10(1):743.
- [Ryckewaert et al., 2023b] Ryckewaert, M., Hérán, D., Trani, J.-P., Mas-Garcia, S., Feilhes, C., Prezman, F., Serrano, E., and Bendoula, R. (2023b). Hyperspectral images of grapevine leaves including healthy leaves and leaves with biotic and abiotic symptoms. *Scientific Data*, 10(1):743.
- [Sajitha et al., 2024] Sajitha, P., Andrushia, A. D., Anand, N., Naser, M., and Lubloy, E. (2024). A deep learning approach to detect diseases in pomegranate fruits via hybrid optimal attention capsule network. *Ecological Informatics*, 84:102859.
- [Sarić et al., 2022] Sarić, R., Nguyen, V. D., Burge, T., Berkowitz, O., Trtílek, M., Whelan, J., Lewsey, M. G., and Čustović, E. (2022). Applications of hyperspectral imaging in plant phenotyping. *Trends in plant science*, 27(3):301–315.
- [Scheibenreif et al., 2023] Scheibenreif, L., Mommert, M., and Borth, D. (2023). Masked vision transformers for hyperspectral image classification. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2166–2176.
- [Shawon et al., 2024] Shawon, S. M., Ema, F. B., Mahi, A. K., Niha, F. L., and Zubair, H. (2024). Crop yield prediction using machine learning: An extensive and systematic literature review. *Smart Agricultural Technology*, page 100718.
- [Slavkovikj et al., 2015] Slavkovikj, V., Verstockt, S., De Neve, W., Van Hoecke, S., and Van de Walle, R. (2015). Hyperspectral image classification with convolutional neural networks. In *Proceedings of the 23rd ACM international conference on Multimedia*, pages 1159–1162.
- [Thenkabail et al., 2013] Thenkabail, P. S., Mariotto, I., Gumma, M. K., Middleton, E. M., Landis, D. R., and Huemmrich, K. F. (2013). Selection of hyperspectral narrowbands (hnbs) and composition of hyperspectral twoband vegetation indices (hvis) for biophysical characterization and discrimination of crop types using field reflectance and hyperion/eo-1 data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 6(2):427–439.

- [Thenkabail et al., 2000] Thenkabail, P. S., Smith, R. B., and De Pauw, E. (2000). Hyperspectral vegetation indices and their relationships with agricultural crop characteristics. *Remote sensing of Environment*, 71(2):158–182.
- [Van Delden et al., 2021] Van Delden, S., SharathKumar, M., Butturini, M., Graamans, L., Heuvelink, E., Kacira, M., Kaiser, E., Klamer, R., Klerkx, L., Kootstra, G., et al. (2021). Current status and future challenges in implementing and upscaling vertical farming systems. *Nature Food*, 2(12):944–956.
- [Vaswani et al., 2017] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A., Kaiser, L., and Polosukhin, I. (2017). Attention is all you need. In *Advances in neural information processing systems*.
- [Verma et al., 2024] Verma, T., Wei, N. T. Z., Gao, F., Yu, H., and Shirota Filho, R. (2024). Optimizing indoor farming: Deep learning for predicting plant growth under led light treatments. In *2024 IEEE Conference on Artificial Intelligence (CAI)*, pages 1051–1056. IEEE.
- [Xu et al., 2023] Xu, P., Fu, L., Xu, K., Sun, W., Tan, Q., Zhang, Y., Zha, X., and Yang, R. (2023). Investigation into maize seed disease identification based on deep learning and multi-source spectral information fusion techniques. *Journal of Food Composition and Analysis*, 119:105254.
- [Yang et al., 2021] Yang, W., Nigon, T., Hao, Z., Paiao, G. D., Fernández, F. G., Mulla, D., and Yang, C. (2021). Estimation of corn yield based on hyperspectral imagery and convolutional neural network. *Computers and Electronics in Agriculture*, 184:106092.
- [Zhang et al., 2023a] Zhang, M., Zhang, C., Zhang, Q., Guo, J., Gao, X., and Zhang, J. (2023a). Es-saformer: Efficient transformer for hyperspectral image super-resolution. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 23073–23084.
- [Zhang et al., 2023b] Zhang, S., Jiang, Y., Wang, C., Liu, W., Li, D., and Wu, K. (2023b). Crop type classification network for hyperspectral images coupled with agricultural vegetation indices. In *2023 11th International Conference on Agro-Geoinformatics (Agro-Geoinformatics)*, pages 1–6. IEEE.
- [Zhao et al., 2023a] Zhao, C., Qin, B., Feng, S., Zhu, W., Sun, W., Li, W., and Jia, X. (2023a). Hyperspectral image classification with multi-attention transformer and adaptive superpixel segmentation-based active learning. *IEEE Transactions on Image Processing*, 32:3606–3621.
- [Zhao et al., 2023b] Zhao, M., Cang, H., Chen, H., Zhang, C., Yan, T., Zhang, Y., Gao, P., and Xu, W. (2023b). Determination of quality and maturity of processing tomatoes using near-infrared hyperspectral imaging with interpretable machine learning methods. *LWT*, 183:114861.
- [Zhu et al., 2024] Zhu, Z., Wang, X., Li, G., and Zhong, Y. (2024). A self-supervised spaceborne multispectral and hyperspectral image fusion unrolling network. *IEEE Transactions on Geoscience and Remote Sensing*.